

Adam: Momentum and RMSProp Together

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** [Optimizers in deep learning](#); [Exponentially weighted moving average \(EWMA\)](#); [SGD with momentum](#); [RMSProp](#).
- **Leads to:** [Learning-rate warm-up schedule](#).
- **Compare with:** [RMSProp](#).

1. Overview

Key point: Adam keeps two running averages (EWMAs) per parameter: of the gradient (momentum's idea) and of the squared gradient (RMSProp's idea). It corrects both for starting at 0, then steps by the first divided by the root of the second. It is the most widely used optimizer and the usual starting point.

Adam (G-169), short for *adaptive moment estimation* (Kingma and Ba 2015), is the last **optimizer** (G-1401) of this series and the most used. Whether we train a plain network, a **convolutional network** (G-484; a network for images) or a **recurrent network** (G-1647; a network for sequences such as text), Adam is usually the first choice.

Adam borrows from the optimizers before it. **Momentum** (G-1258) and **NAG** (G-1315) are built on one idea, **speed from past gradients**; **AdaGrad** (G-168) and **RMSProp** (G-1697) on another, **a learning rate that adapts to each parameter**. Adam merges the two.

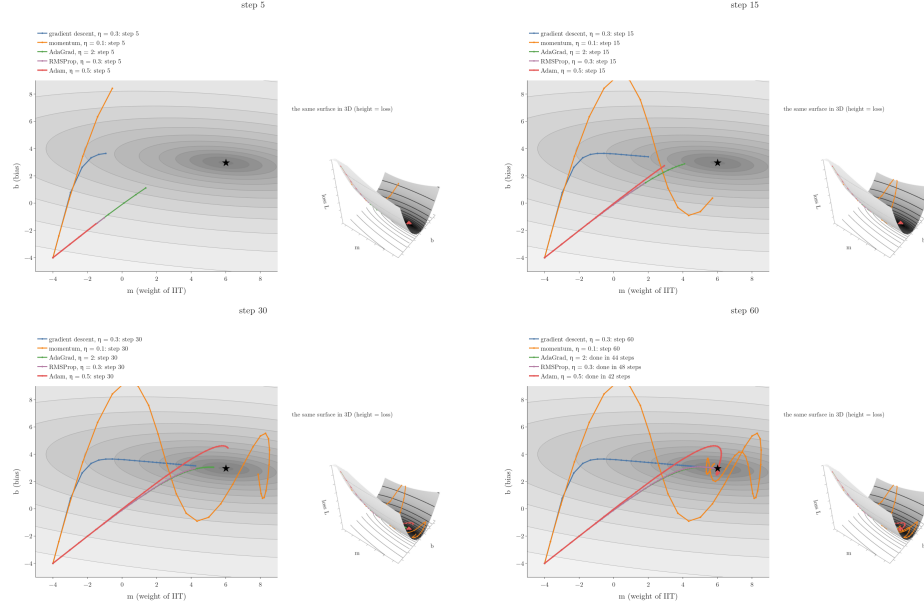


Figure 1: Five optimizers on the elongated bowl of a sparse feature, each with a learning rate that works for it. Adam (red) heads towards the minimum like the adaptive methods and curls in like momentum

Figure 1 is a **contour map** (the surface seen from above; each line joins points of equal loss; [how to read a contour map](#)), with the same surface in 3D at its right (height = the loss; the lines on it are the lines of the map). The surface is the loss of the students data, a bowl stretched along m (built, and tilted from the side to the top view, in [a sparse feature stretches the loss](#)):

$$L(m, b) = \frac{1}{100} \sum_{i=1}^{100} (y_i - (m x_i + b))^2$$

At the start of the paths:

$$L(-4, -4) = 72.63$$

Here x_i is the IIT value (0 or 1) of student i and y_i their package.

Lines close together mean steep (across the bowl, along b), lines far apart mean flat (along m), and the star is the lowest point, $L(6.01, 2.96) = 0.23$.

Figure 1 shows both ideas at work on the students data.

2. Prerequisites

- [Momentum's velocity](#), a running average of past gradients.
- [RMSProp's division by the root of an EWMA of squared gradients](#).

- Why an average started at 0 is too small at first.

3. The story so far

Key point: Batch GD is steady but slow; momentum is fast but overshoots; NAG damps the overshoot; AdaGrad handles sparse features but its learning rate dies; RMSProp fixes that.

Optimizer	What it adds	Its problem
Batch gradient descent	the basic step	slow
Momentum	speed from past gradients	overshoots and oscillates
NAG	look ahead, damp the oscillations	can stay in small dips
AdaGrad	a learning rate per parameter (sparse features)	the learning rate only shrinks
RMSProp	forget old gradients	no correction for starting at 0

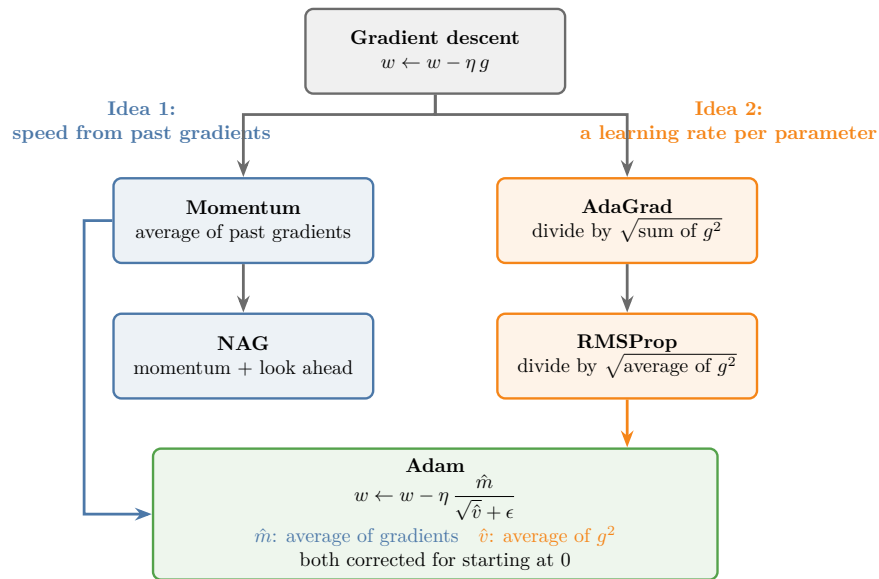


Figure 2: The two lines of optimizers. Left (blue): speed from past gradients. Right (orange): a learning rate per parameter. Adam takes the average of gradients from the left line and the average of squared gradients from the right line

Figure 2 draws the table as a family tree. Follow the two coloured arrows into Adam: the blue one brings momentum’s average of past gradients, the orange one brings RMSProp’s average of squared gradients.

The two lines of improvement are independent, so it makes sense to combine them. Adam

is perhaps best seen as a combination of RMSProp and momentum, with a few important differences (Goodfellow et al. 2016, §8.5.3).

4. The update rule

Key point: Three lines per step: update the two averages, correct both for starting at 0, then move by the corrected average gradient divided by the root of the corrected average squared gradient.

1. In words:

- keep an EWMA (**exponentially weighted moving average**, G-735) of the gradient, m_t , as momentum does;
- keep an EWMA of the squared gradient, v_t , as RMSProp does;
- correct both for having started at 0;
- step in the direction of the corrected average gradient, divided by the root of the corrected average squared gradient.

2. Formula:

$$\begin{aligned}
 m_t &= \beta_1 m_{t-1} + (1 - \beta_1) \nabla L(w_t) \\
 v_t &= \beta_2 v_{t-1} + (1 - \beta_2) (\nabla L(w_t))^2 \\
 \hat{m}_t &= \frac{m_t}{1 - \beta_1^t} \\
 \hat{v}_t &= \frac{v_t}{1 - \beta_2^t} \\
 w_{t+1} &= w_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t
 \end{aligned}$$

with $m_0 = v_0 = 0$, where t counts the updates: 1 for the first mini-batch, 2 for the second, and so on. The suggested defaults are $\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 10^{-8}$ (Kingma and Ba 2015, algorithm 1).

3. Example: one weight with gradients 2 and then 1, using the defaults.

Step $t = 1$, gradient 2:

$$\begin{aligned}
 m_1 &= 0.1 \times 2 = 0.2 \\
 v_1 &= 0.001 \times 2^2 = 0.004 \\
 \hat{m}_1 &= 0.2 / 0.1 = 2 \\
 \hat{v}_1 &= 0.004 / 0.001 = 4 \\
 \text{step} &= 0.001 \times 2 / \sqrt{4} \\
 &= 0.001
 \end{aligned}$$

Step $t = 2$, gradient 1:

$$m_2 = 0.9 \times 0.2 + 0.1 \times 1$$

$$\begin{aligned}
&= 0.28 \\
v_2 &= 0.999 \times 0.004 + 0.001 \times 1^2 \\
&= 0.004996 \\
\hat{m}_2 &= 0.28 / (1 - 0.9^2) = 0.28 / 0.19 \\
&= 1.474 \\
\hat{v}_2 &= 0.004996 / (1 - 0.999^2) \\
&= 0.004996 / 0.001999 = 2.499 \\
\text{step} &= 0.001 \times 1.474 / \sqrt{2.499} \\
&= 0.000932
\end{aligned}$$

(Notebook.)

The parts come from the earlier optimizers:

- m_t is momentum's velocity, written as a true EWMA. In Adam it is an estimate of the **first moment** (G-780; the mean) of the gradient (Goodfellow et al. 2016, §8.5.3).
- $\sqrt{v_t}$ in the denominator is RMSProp's. v_t estimates the **second moment** (G-1759; the mean of the square) of the gradient, which gives Adam its name: adaptive **moment** estimation.
- $\hat{m}_t$ and $\hat{v}_t$ are new: the **bias correction** (G-285).

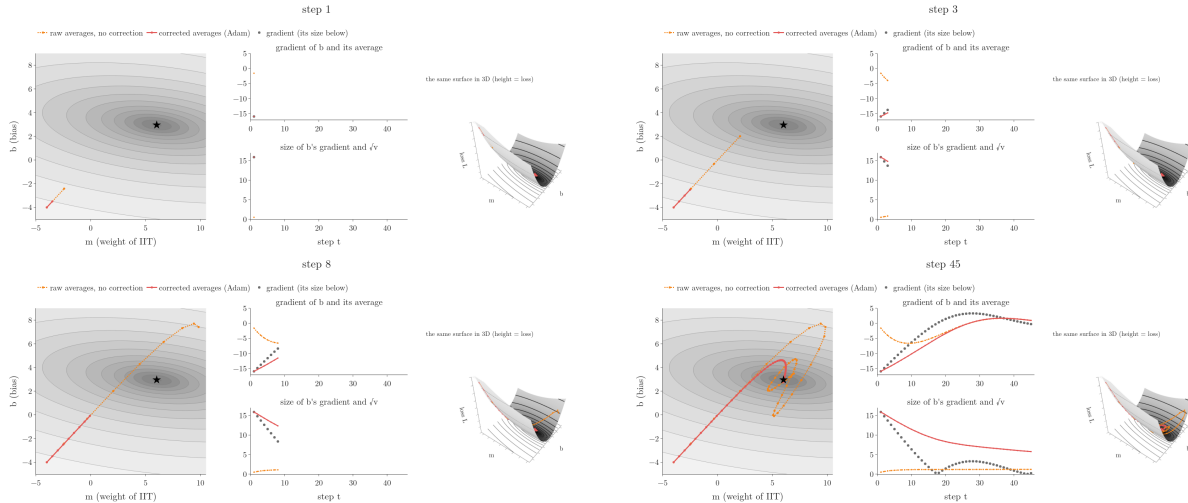


Figure 3: Adam, $\eta = 0.5$, on the students data, step by step. Left: the path with corrected averages (red) and with the raw ones (dotted orange). Right, for the bias b of the red run: the gradient (grey dots) with m raw and corrected (top), and the gradient's size with $\sqrt{v}$ raw and corrected (bottom)

Figure 3 runs the rule on the students data of Figure 1. Watch the right panels: the raw averages start near 0 (at step 1, $\sqrt{v}$ is 0.50 while the gradient's size is 15.9), and the corrected ones start on the gradient. With the raw averages the first step is 2.24 long instead of 0.71, and the path overshoots to $b = 7.7$ before it comes back; section 5 explains why.

5. Bias correction

Key point: An EWMA started at 0 is too small at first: after t steps its weights add up to only $1 - \beta^t$ (0.1 after one step with $\beta = 0.9$). Dividing by $1 - \beta^t$ removes this start-up bias. With $\beta_2 = 0.999$ the bias would last for thousands of steps.

Both averages start at $m_0 = v_0 = 0$, so in the first steps they are pulled towards 0, exactly like [the zero start of an EWMA](#).

How big the bias is. Unrolled, the EWMA after t steps is a weighted sum of the gradients $g_1, \dots, g_t$, with weight $(1 - \beta)\beta^{t-i}$ on gradient g_i :

$$(1 - \beta) \sum_{i=1}^t \beta^{t-i} g_i$$

If every gradient had the same value g , this would be

$$\begin{aligned} (1 - \beta) (1 + \beta + \dots + \beta^{t-1}) g \\ = (1 - \beta^t) g \end{aligned}$$

because the geometric sum $1 + \beta + \dots + \beta^{t-1}$ equals $(1 - \beta^t)/(1 - \beta)$. Check with $\beta = 0.9$ and $t = 3$:

$$\begin{aligned} 1 + 0.9 + 0.81 &= 2.71 \\ \frac{1 - 0.9^3}{1 - 0.9} &= \frac{1 - 0.729}{0.1} = 2.71 \end{aligned}$$

The average is too small by exactly the factor $1 - \beta^t$, so dividing by it gives back g (Kingma and Ba 2015, §3).

	$t = 1$	$t = 10$	$t = 100$	$t = 1000$
$1 - 0.9^t$	0.1	0.651	1.000	1.000
$1 - 0.999^t$	0.001	0.010	0.095	0.632

With $\beta_1 = 0.9$ the factor reaches 1 within a few dozen steps; with $\beta_2 = 0.999$ it is still 0.63 after 1,000 steps (Notebook). The larger β is, the longer the bias lasts (Kingma and Ba 2015, §2).

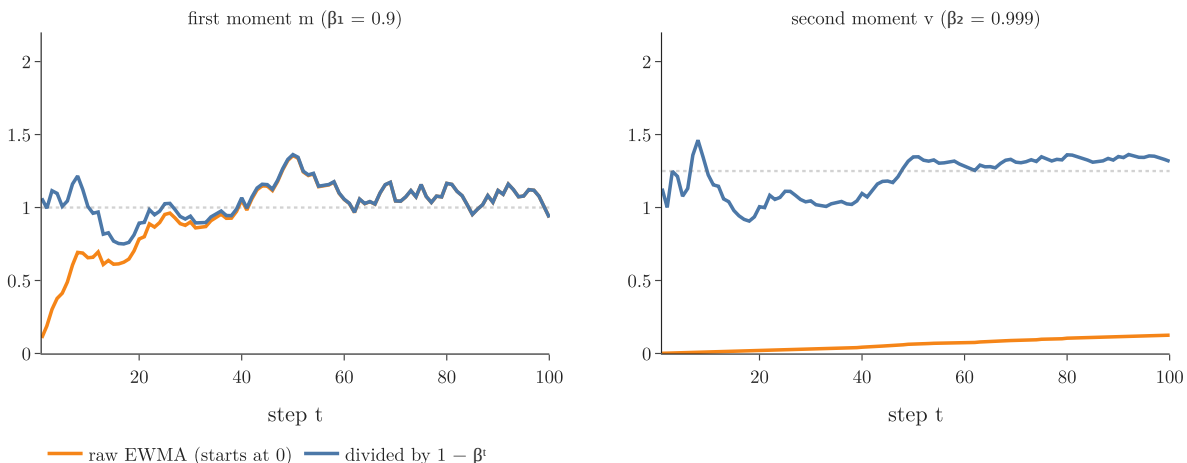


Figure 4: EWMA of a noisy gradient with mean 1 (true second moment 1.25), started at 0. The raw averages (orange) start near 0 and climb slowly, very slowly for $\beta_2 = 0.999$; the corrected ones (blue) are on target from the first step

Figure 4 shows both averages on a noisy gradient. At step 10, the raw v is 0.012 against a true value of 1.25; corrected, it is 1.22 (Notebook).

Without the correction, early steps would be badly scaled: at $t = 1$ in the example above, the raw ratio is

$$\begin{aligned} m_1/\sqrt{v_1} &= 0.2/\sqrt{0.004} \\ &= 3.16 \end{aligned}$$

instead of the corrected 1, a first step more than three times too large. Kingma and Ba (2015, §3) point out that leaving the correction out leads to much larger initial steps. RMSProp keeps an uncorrected second-moment estimate, which may be strongly biased early in training (Goodfellow et al. 2016, §8.5.3).

6. Adam on the students data

Key point: Adam heads for the minimum like an adaptive method, curls in like momentum, and then settles, while RMSProp keeps jittering.

On the **elongated bowl** (G-673) of the IIT feature (Figure 1), each optimizer gets a learning rate that works for it. Steps until the loss is within 0.01 of its minimum (Notebook):

Optimizer	η	Steps	Still moving at the end?
Gradient descent	0.3	61	no
Momentum, $\beta = 0.9$	0.1	64	no
AdaGrad	2	44	no
RMSProp, $\beta = 0.9$	0.3	48	yes: up to 0.15 from the best
Adam	0.5	42	no

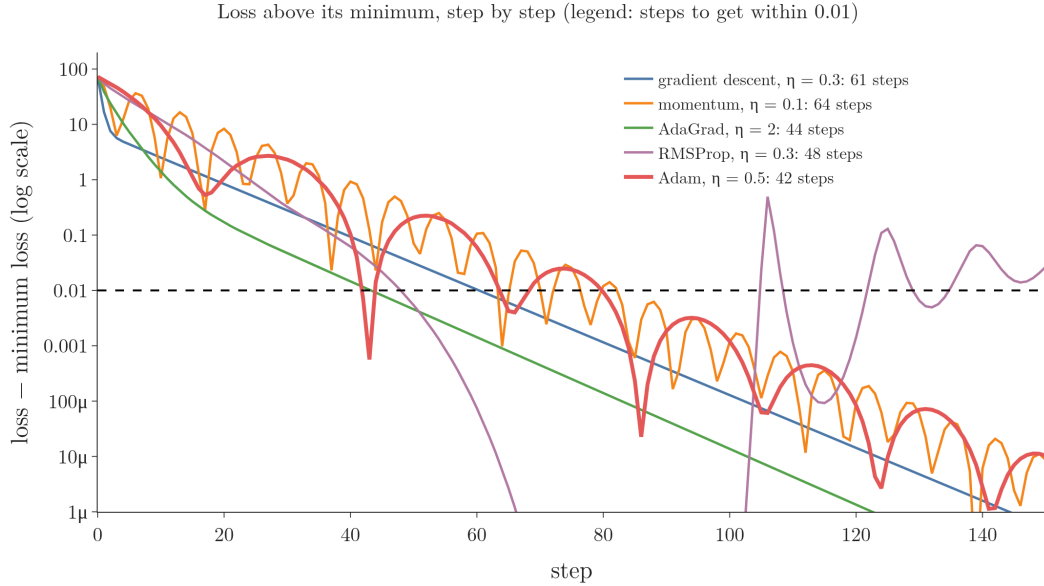


Figure 5: The loss above its minimum at every step for the five runs of the table, on a log scale. The dashed line is the 0.01 target; the legend gives the step at which each run first gets below it.

In Figure 5 the vertical axis is how far the loss is above its lowest possible value, on a log scale: each labelled line is 10 times the one below it (μ means one millionth, so 100μ is 0.0001), so the lower a curve, the closer the run is to the minimum. The dashed line is the 0.01 target.

Figure 5 shows the same runs step by step. The step counts in the table are the first time a run gets within 0.01; the curves show what happens next. Gradient descent and AdaGrad keep falling smoothly. Momentum and Adam cross the line, swing back above it, and settle in smaller and smaller swings. RMSProp falls steadily, then very steeply after step 45, then jumps back up near step 105 and keeps bouncing around 0.01 to the end.

Momentum is no faster than gradient descent here (64 steps against 61). Its speed carries it past the minimum on the way down, so its loss rises and falls several times before it first gets within 0.01 (the bumps in its curve in Figure 5). These are the overshoot and oscillation of the table in section 3; on a small convex bowl that a plain step already crosses quickly, the swings cost the steps the speed gained.

Adam’s path shows both behaviours. Adam moves in m and b together from the start, like AdaGrad and RMSProp, instead of the “L” of gradient descent. Near the minimum it swings around a few times, each swing smaller, the momentum part, and then settles. RMSProp, with steps of about η even near the minimum, keeps jittering up to 0.15 away. On a convex bowl like this one (a **convex function**, G-476) the differences are small; Adam’s strengths matter most on the complex, non-convex losses of real networks (**non-convex functions**, G-1333).

7. Adam on real data: MNIST

Key point: On MNIST, Adam and RMSProp train fastest and are close to each other; both are far ahead of plain SGD and AdaGrad, and ahead of momentum and NAG.

The setup:

- **data:** the **MNIST** (G-1249) handwritten digits ([the MNIST data](#)), 10,000 training images, each with 784 pixel **features** (G-772; input variables) and the digit as **target** (G-1949; the output we predict), and the 10,000 test images for validation (checking accuracy on images the network did not train on);
- **network:** hidden layers of 128 and 64 **ReLU** (G-1668; each node outputs its input if positive, else 0) nodes and a softmax output (10 probabilities, one per digit);
- **training:** **batch size** (G-267) 64, 20 **epochs** (G-696), 3 seeds;
- **learning rates** (G-1068): 0.01 for SGD, momentum, NAG ($\beta = 0.9$) and AdaGrad; 0.001, Keras’ default, for RMSProp and Adam.

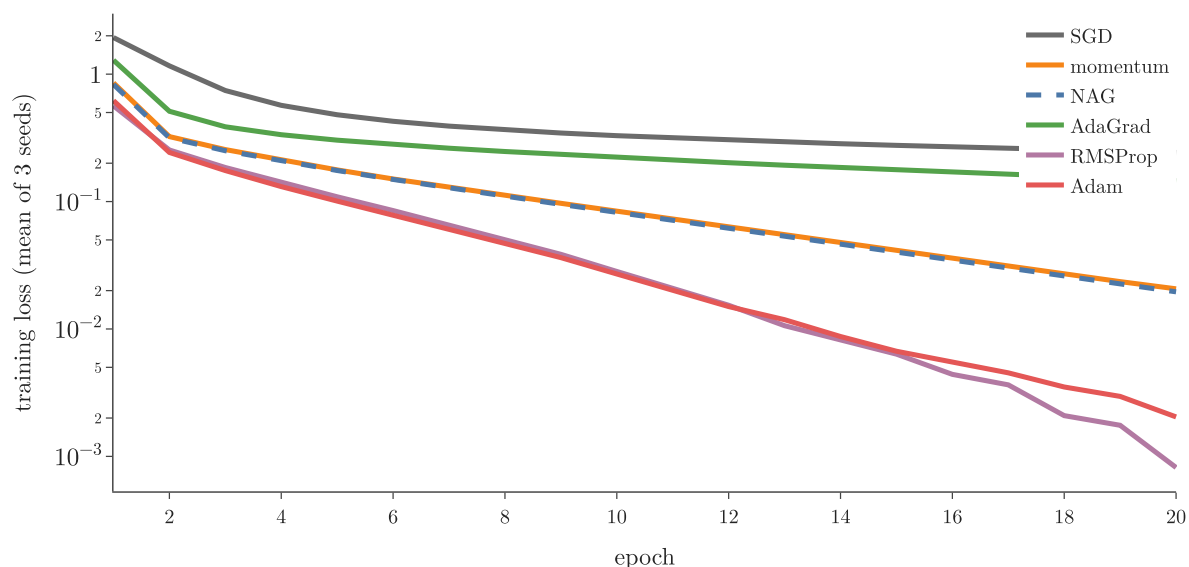


Figure 6: Training loss on MNIST per epoch for six optimizers, mean of 3 seeds

In Figure 6 each curve is the training loss after each epoch, on a log scale (each labelled power of ten is 10 times the one below; the small numbers between mark 2, 3, ... times it).

A lower curve is better.

Figure 6 and the Notebook give, as means over 3 seeds:

	SGD	Momentum	NAG	AdaGrad	RMSProp	Adam
Learning rate	0.01	0.01	0.01	0.01	0.001	0.001
Training loss, epoch 1	1.94	0.85	0.83	1.29	0.56	0.62
Training loss, epoch 5	0.48	0.18	0.18	0.30	0.11	0.10
Training loss, epoch 20	0.24	0.021	0.020	0.15	0.0008	0.0020
Validation accuracy, epoch 20	0.917	0.949	0.949	0.934	0.956	0.954

Adam behaves like both of its parents. Adam gets the speed of momentum and the per-parameter learning rates of RMSProp, so it is far ahead of plain SGD and AdaGrad, and ahead of momentum and NAG from the first epochs on. Against RMSProp alone the race is close on this small network: Adam is slightly lower by epoch 5, RMSProp slightly lower by epoch 20, and their validation accuracies differ by about 0.001 (0.9556 against 0.9542; Notebook). The learning rates are each optimizer’s usual value, not tuned, so small differences between the leaders should not be over-read; Adam’s advantage is that it does well across many problems with these defaults (section 8).

8. Which optimizer to use

Key point: Start with Adam. If the results are not good, try RMSProp, and sometimes momentum. There is no single best optimizer: tune the choice like any hyperparameter.

There are more optimizers than the five covered here, such as AdaDelta and Nadam (Adam with Nesterov momentum). Which one to use has no clear answer: a large comparison across many tasks found the adaptive methods fairly robust, but no single best algorithm (Goodfellow et al. 2016, §8.5.4). A practical order:

1. **Adam** is a good starting point; over recent years it has given good results on many kinds of problems.
2. If Adam’s results are not good, try **RMSProp**; sometimes **momentum** works best.
3. Treat the optimizer as a **hyperparameter** (G-910; a setting chosen before training; [Bayesian search with Optuna](#) covers tuning in general): try several and keep the one

that does best on validation data, for example with **Keras Tuner** (G-1001; [choosing the optimizer with Keras Tuner](#)).

Adam is generally regarded as fairly robust to the choice of its hyperparameters, though the learning rate sometimes needs changing from the suggested default of 0.001 (Goodfellow et al. 2016, §8.5.3). Its authors describe the hyperparameters as having intuitive meanings and typically requiring little tuning (Kingma and Ba 2015).

Extra: Ruder (2016, §4.10) recommends the adaptive methods when the data is sparse or the network deep or complex, and notes that Kingma and Ba found the bias correction helps Adam slightly outperform RMSProp towards the end of optimisation, as gradients become sparser.

9. Adam in Keras

Key point: `keras.optimizers.Adam(learning_rate=0.001, beta_1=0.9, beta_2=0.999)`, or simply `optimizer="adam"`.

Python: Adam in Keras.

```
import keras
opt = keras.optimizers.Adam(learning_rate=0.001, beta_1=0.9,
                             beta_2=0.999)
model.compile(optimizer=opt, loss="sparse_categorical_crossentropy",
              metrics=["accuracy"])
```

The defaults match the paper's except `epsilon=1e-7` (paper: 10^{-8}) (Keras Adam documentation).

10. Summary

	Momentum	RMSProp	Adam
EWMA of gradients	yes (v_t)	no	yes (m_t , $\beta_1 = 0.9$)
EWMA of squared gradients	no	yes ($\beta = 0.9$)	yes (v_t , $\beta_2 = 0.999$)
Learning rate per parameter	no	yes	yes
Bias correction	no	no	yes, divide by $1 - \beta^t$
Default learning rate in Keras	0.01 (SGD)	0.001	0.001

- Adam = momentum’s average of gradients + RMSProp’s average of squared gradients + a correction for starting at 0.
- The update divides the corrected average gradient by the root of the corrected average squared gradient (the formula of section 4).
- Defaults: $\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$; t counts the updates.
- Adam is the usual first choice; RMSProp and momentum are the usual alternatives. No optimizer wins everywhere.

11. Sources

Built from

- CampusX, “Adam Optimizer Explained in Detail with Animations | Optimizers in Deep Learning Part 5”, YouTube, <https://www.youtube.com/watch?v=N5AynalXD9g>

Other references

- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*. MIT Press. §8.5.3 Adam (algorithm 8.7), §8.5.4 Choosing the right optimization algorithm.
- Kingma, D. P. and Ba, J. (2015). Adam: A method for stochastic optimization. ICLR 2015. arXiv:1412.6980. Algorithm 1, §2, §3.
- Ruder, S. (2016). An overview of gradient descent optimization algorithms. arXiv:1609.04747. §4.6 Adam, §4.10 Which optimizer to use?
- Keras API documentation: Adam optimizer, keras.io/api/optimizers/adam.

12. Key terms

Term	Meaning
Adam	Adaptive moment estimation: an optimizer that combines momentum’s EWMA of gradients with RMSProp’s EWMA of squared gradients, plus bias correction

Term	Meaning
First moment m_t	The EWMA of the gradient: an estimate of its mean
Second moment v_t	The EWMA of the squared gradient: an estimate of its mean square
Bias correction	Dividing an EWMA started at 0 by $1 - \beta^t$ so it is not too small in the first steps
β_1, β_2	Adam's decay factors for m_t and v_t ; defaults 0.9 and 0.999
Hyperparameter	A setting chosen before training, such as the optimizer or the learning rate
Feature	An input variable, such as one pixel of an image
Target	The output we predict, such as the digit